

Recent Kolmogorov–Arnold Network (KAN) Architecture -level developments

Purpose and Scope

Note : no paper using KAN in single-molecule localization microscopy, or PSF-based emitter localization was found in this window. This is consistent with the KAN-SMLM direction remaining a novel, unclaimed intersection as of this search.

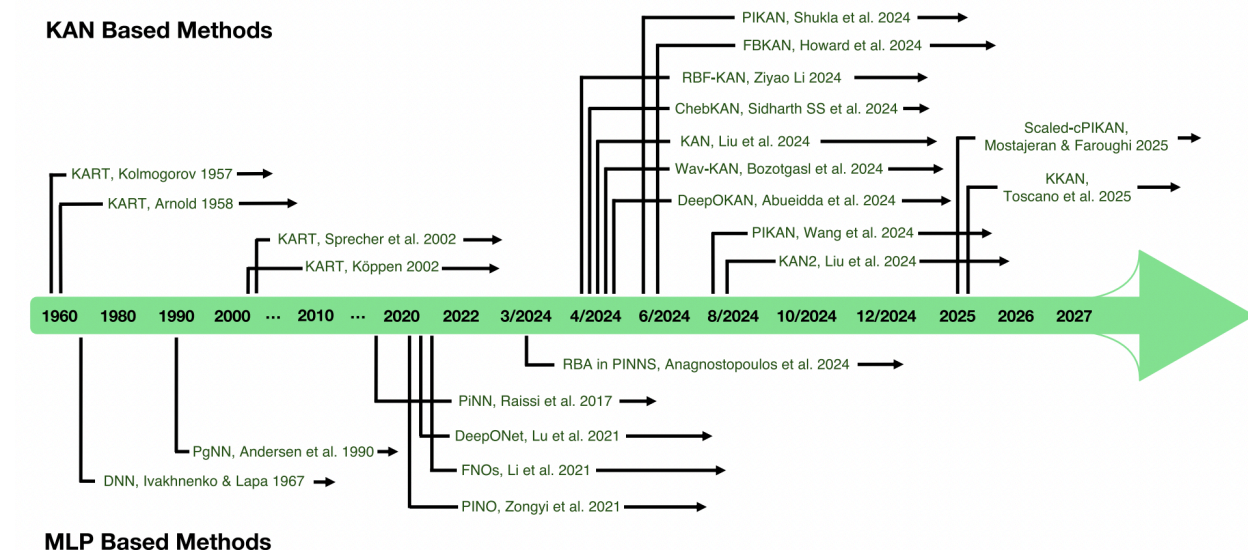
1. Faroughi et al. — KAN Review, Synthesis, and New Analysis

Neural Networks 200 (2026) 108791

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Abstract

A large review-and-synthesis article organizing KAN research into three modeling paradigms: data-driven KANs, physics-informed KANs (PIKANs), and deep-operator KANs (DeepOKANs). Alongside the literature review, the authors ran their own new comparative experiments benchmarking KAN-family models against MLPs, PINNs, and DeepONets on accuracy, convergence, and spectral (Neural Tangent Kernel) behavior.



Core theory recap

The Kolmogorov-Arnold Representation Theorem (KART) states that for continuous $f: [0, 1]^n \rightarrow \mathbb{R}$, there exist continuous univariate functions φ_k and $\psi_{k,j}$ such that:

$$f(x_1, x_2, \dots, x_n) = \sum_{k=0}^{2n} \varphi_k \left(\sum_{j=1}^n \psi_{k,j}(x_j) \right) \quad \text{for all } (x_1, \dots, x_n) \in [0, 1]^n.$$

A two-layer KAN implements this directly: layer 1 applies $\psi_{\square, \square}$ to each input dimension and sums over j ; layer 2 applies $\phi_{\square}$ to that sum and sums over k . Multi-layer KANs generalize this by nesting univariate compositions, $\phi_{\square}(\cdot) = \phi_{\square}^{(L)}(\dots \phi_{\square}^{(2)}(\phi_{\square}^{(1)}(\cdot))\dots)$, a notion the paper calls "Kolmogorov-Arnold depth," following Liu et al.

The paper gives an explicit approximation-rate comparison between KAN and MLP under smoothness assumptions:

$$\|y - y_{\text{KAN}}\|_{\infty} \leq C_{\text{KAN}}(y, n) \cdot |\theta|^{-\alpha}, \quad \|y - y_{\text{MLP}}\|_{\infty} \leq C_{\text{MLP}}(y, n) \cdot |\theta|^{-\beta},$$

with $\alpha > \beta > 0$

for a target y with parameter counts $|\theta|$. KANs are stated to have a faster asymptotic approximation rate exponent than MLPs under these assumptions, meaning more efficient use of parameters when the target function has dimension-wise structure exploitable by univariate transformations.

Basis-function options reviewed, with parameter-count order ($N_{\square}$ = layers, $N_{\square}$ = neurons/layer, g = grid size, k = polynomial degree):

- B-splines (original KAN): $\phi(\xi) = w_{\square} b_{\square}(\xi) + w_{\square} \sum_i c_i B_i(\xi)$, with $b(\xi) = \xi / (1 + e^{-\xi})$ (SiLU). Parameter order $O(N_{\square} N_{\square}^2 (k+g))$.
- Chebyshev polynomials: $\phi(\xi) = \sum_i c_i T_i(\xi)$. Parameter order $O(N_{\square} N_{\square}^2 k)$ — fewer parameters, fast convergence for smooth functions, but requires input normalization to $[-1, 1]$ (via $\tanh$ after each layer) to avoid NaN losses in the trigonometric evaluation form $T_i(\xi) = \cos(i \cdot \arccos(\xi))$.
- Wavelets (Wav-KAN): $\phi(\xi) = \sum_{\ell} \sum_r d_{\ell, r} \zeta_{\ell, r}(\xi)$, parameter order $O(3N_{\square} N_{\square}^2)$; good for localized/high-frequency structure.
- Radial basis functions (RBF / FastKAN): $\phi(\xi) = \sum_r w_r \kappa((\xi - c_r)/\sigma_r)$, parameter order $O(3N_{\square} N_{\square}^2)$; smooth interpolation, faster per-step than B-splines.

Data-driven KANs — architecture and reported comparisons

Training minimizes MSE loss $L_{\text{Data}} = (1/N) \sum_i (y_{\text{KAN}}(\xi_i; \theta) - y_i)^2$ via backpropagation, typically Adam, sometimes combined with L-BFGS for late-stage refinement.

Comparative findings reported by the authors' own new analysis (Mostajeran & Faroughi elasto-plasticity case study, Figs. 5–6 in the source):

- Accuracy: Chebyshev-KAN (cKAN) architectures (serial and parallel) showed substantially lower absolute prediction errors than size-matched MLPs for stress components p and q under an undrained triaxial loading path ($\xi = -1000$); e.g. serial cKAN 0.98% error vs. serial MLP 6.69% for p , and 1.76% vs. 5.94% for q .
- Convergence: cKAN models showed gradual, sustained cost-function decrease during training, while MLPs converged faster initially but plateaued early ("premature stagnation"), particularly in the serial architecture.
- Per-epoch wall-clock cost (2000 epochs, same hardware): serial cKAN ≈ 345 ms/epoch and parallel cKAN ≈ 337 ms/epoch, versus serial MLP ≈ 204 ms/epoch and parallel MLP ≈ 212 ms/epoch — i.e. cKAN was reported as slower per epoch but needed fewer epochs to reach a given error level.
- Neural Tangent Kernel (NTK) spectral analysis: cKAN exhibited a more stable, compact eigenvalue spectrum during training than MLP, which the authors link to better conditioning and more consistent generalization; this held across 10 random initialization seeds with narrow standard-deviation bands.

Physics-informed KANs (PIKANs)

PIKAN loss combines data, PDE-residual, initial-condition, and boundary-condition terms:

$$L = w_{\text{Data}} L_{\text{Data}} + w_{\text{PDE}} L_{\text{PDE}} + w_{\text{IC}} L_{\text{IC}} + w_{\text{BC}} L_{\text{BC}}$$

with $L_{PDE} = \sum_i \| [u_KAN(\xi_i^*r; \theta)] - f(\xi_i^*r) \|^2$ evaluated at residual collocation points, and analogous forms for L_{IC} and L_{BC} . Constraints may be soft (penalty terms, as above) or hard (re-parameterizing the output as $u_\theta(\xi) = g(\xi) + h(\xi) \cdot u_KAN(\xi; \theta)$ so boundary/initial conditions are satisfied by construction).

Reported accuracy comparison on the Allen-Cahn equation (Table 5 in the source, collecting results from Shukla et al. 2024, Mostajeran & Faroughi 2025, Toscano et al. 2025c, and Kiyani et al. 2025c):

Configuration	Parameters $ \theta $	Relative L^2 error
PIKAN, B-spline, Adam	6721	5.83×10^{-1}
PIKAN, Chebyshev, Adam	6720	5.15×10^{-2}
PINN, tanh, Adam	7850	5.10×10^{-1}
PIKAN, Chebyshev, Adam+BFGS	1368	1.23×10^{-3}
PIKAN, Chebyshev, Adam+SSBroyden	1368	9.01×10^{-6}
PINN, tanh, Adam+SSBroyden	2019	1.15×10^{-6}

The paper reports that switching PIKAN's basis from B-spline to Chebyshev cut error by over 90% at equal parameter count ($5.83 \times 10^{-1} \rightarrow 5.15 \times 10^{-2}$), and that pairing Chebyshev-PIKAN with the SSBroyden optimizer gave a 99.3% error reduction versus the same architecture with Adam/BFGS. Across the broader comparison table, PIKAN is reported to outperform PINN by 85–99% in relative error in most configurations, generally with fewer trainable parameters.

A domain/residual scaling method ("Scaled-cPIKAN," from the authors' own prior work) is reported to achieve an 88% error reduction versus Scaled-PINN and a 90% improvement over vanilla cPIKAN when solving Allen-Cahn over an extended domain $[-6, 6]$, by mapping inputs and residuals into the interval natural to the Chebyshev basis.

NTK/spectral-entropy analysis: the paper defines spectral entropy of the NTK eigenvalue spectrum as

$$\text{Spectral Entropy} = - \sum_i (|\lambda_i| / \sum |\lambda|) \cdot \log(|\lambda_i| / \sum |\lambda|)$$

and reports that cPIKAN consistently achieves lower spectral entropy than PINN on a forced-vibration benchmark with higher-order derivative terms (u_{xxxx} , u_{tt}), interpreted as better alignment with high-frequency solution content.

Deep-operator KANs (DeepOKANs)

Structured like DeepONet (branch + trunk networks combined via inner product), but both subnetworks are KANs rather than MLPs:

$$G_{\theta}(v)(xi) = \text{Sum}_{k=1..q} [\text{beta}_k(v; \theta_b) * \text{tau}_k(xi; \theta_t)]$$

Reported accuracy comparison on 1D Darcy flow (Pant et al. 2025 results, reproduced in the source's Table 7): in shallow architectures, DeepOKAN (B-spline basis) reduced relative L^2 error from DeepONet's 2.89% to 0.39% (0% noise) — over 85% improvement; in deep architectures the gap narrowed but DeepOKAN still improved error by roughly 65%. Under added input noise (5–10%), DeepOKAN's advantage persisted but narrowed to single digits of percentage improvement.

On the 2D Darcy flow problem (Shukla et al. 2024 results): DeepONet had lower error at 0% noise (1.62% vs. DeepOKAN's 2.18%) but degraded faster with added noise, while DeepOKAN's error stayed nearly flat (2.18% $\rightarrow$ 2.30% from 0% to 10% noise), reported as evidence of better noise robustness for the KAN-based operator.

Convergence: DeepOKAN is reported to reach lower training loss with fewer iterations than DeepONet across shallow/deep and low/high-complexity settings, including under different random initialization seeds.

Stated challenges and limitations

- KANs are reported (citing Zeng et al. 2024a) to underperform MLPs on functions with jump discontinuities or non-differentiable points, and to be more sensitive to noise, with training time sometimes exceeding MLP training time "by up to two orders of magnitude."
- Pourkamali-Anaraki (cited) reported KAN performance degrades substantially in low-data regimes.
- KANs lack built-in support in mainstream frameworks (TensorFlow/PyTorch), and the recursive nature of B-splines resists full GPU vectorization — a limitation the paper flags as motivating basis-function alternatives (this is the direct link to the RSS-KAN and Clifford-KAN papers summarized below, though the review itself does not cite either).
- The paper explicitly states no theoretical NTK/generalization analysis yet exists for the broader PIKAN/DeepOKAN family beyond the specific cases studied, and calls this an open research direction.

Relevance to KAN-SMLM work

This paper does not mention microscopy or localization; the following is the report author's inference, not a claim in the paper. The accuracy/parameter-efficiency comparisons for Chebyshev-basis KANs versus B-spline KANs (data-driven and physics-informed) may be relevant context if a basis-function ablation is considered for the KAN-SMLM localizer. The NTK spectral-stability analysis methodology could also be a template for diagnosing training behavior in the DH-PSF Fisher Information work, if such analysis is ever pursued.

2. Zhang, Fan & Lu — RSS-KAN

Neurocomputing 672 (2026) 132796 — Full text reviewed

DOI: <https://doi.org/10.1016/j.neucom.2026.132796>

What the paper proposes

RSS-KAN replaces the B-spline (or single ReLU, as in prior ReLU-KAN) basis function with a learnable hybrid of ReLU, Sigmoid, and Sine components, aiming to combine local support, smooth shape control, and high-frequency/periodic capture in a single basis while remaining fully expressible as matrix operations (no recursion) for GPU efficiency.

The hybrid basis function for edge i is defined as:

$$H_i(x) = [\text{ReLU}(e_i - x) \cdot \text{ReLU}(x - s_i)]^2 \cdot [\alpha^1_i + \alpha^2_i \cdot \sigma(\beta(x - c_i)) + \alpha^3_i \cdot \sin(2\pi(x - s_i)/(e_i - s_i))]]$$

where $\text{ReLU}(x) = \max\{0, x\}$ acts as a windowing/local-support mechanism defining the non-zero interval $[s_i, e_i]$; σ is the sigmoid function providing smooth shape regulation; $c_i = (e_i + s_i)/2$ is the window center; β is a trainable sigmoid slope; and $\alpha^1_i, \alpha^2_i, \alpha^3_i$ are learnable combination weights. Unlike prior KAN basis families, where all basis functions share one fixed bell-shaped profile shifted in position, RSS-KAN's basis shapes are adaptively reshaped during training via the α weights.

Theoretical results (as stated in the paper)

The paper proves three theorems establishing that RSS-KAN's function space contains those of ReLU-KAN, a Sigmoid-only KAN, and a Sine-only KAN as special cases (by zeroing out the other α terms), i.e. $F_{\text{ReLU}} \subseteq F_{\text{RSS}}, F_{\text{Sigmoid}} \subseteq F_{\text{RSS}}, F_{\text{Sin}} \subseteq F_{\text{RSS}}$. It further proves (Theorem 1, via the classical Cybenko/Hornik universal approximation results) that the Sigmoid-only component is dense in $C(K)$ for compact $K \subset \mathbb{R}$; (Theorem 2, via Fourier decomposition and citing Reinhardt et al.'s SineKAN) that adding the Sine component preserves density in $C(K)$ and extends to $L^2(K)$; and (Theorem 3) that the full multivariate RSS-KAN, built on this dense one-dimensional basis family, inherits the multidimensional universal approximation property established for the original KAN by Liu et al.

Computational design

The entire forward/backward computation is expressed as matrix addition, element-wise (Hadamard) products, and a small set of standard nonlinearities — no recursive spline evaluation. This is presented as the source of RSS-KAN's GPU-parallelization advantage over the original B-spline KAN.

Reported experimental results

Fitting-accuracy benchmark against BSRBF-KAN, Efficient-KAN, Fast-KAN, ReLU-KAN, and SineKAN on 8 test functions f_1 – f_8 (mix of univariate/multivariate, periodic, exponential, composite):

- RSS-KAN achieved the lowest MSE on 6 of 8 functions. Reported improvements: 2–3 orders of magnitude over BSRBF-KAN, 2–3 over Efficient-KAN, 2–5 over Fast-KAN, 1–2 over ReLU-KAN, 1–2 over SineKAN.
- Efficient-KAN had marginally lower MSE than RSS-KAN on two smooth functions (f_3 , f_7), attributed to its Gaussian RBF basis being well-suited to smooth/structured patterns; RSS-KAN still outperformed it on oscillatory functions (f_1 , f_6).
- Specific reported MSE values: f_1 — RSS-KAN 9.98×10^{-7} vs. BSRBF-KAN 2.71×10^{-4} ; f_4 — RSS-KAN 5.78×10^{-7} vs. BSRBF-KAN 5.56×10^{-4} ; f_8 (6-dimensional) — RSS-KAN 2.65×10^{-5} vs. next-best ReLU-KAN 1.34×10^{-4} , demonstrating stable high-dimensional generalization.

Training-speed benchmark (Experiment 1, fixed 500 epochs, six functions f_1 – f_6 , NVIDIA Tesla V100): the original KAN was consistently slowest (e.g. 25.34 s CPU / 49.51 s GPU on f_6) due to recursive B-spline computation. Fast-KAN was fastest overall due to simplified Gaussian-RBF computation. RSS-KAN was reported as roughly 2× slower than Fast-KAN/ReLU-KAN, comparable in speed to BSRBF-KAN and Efficient-KAN, but 5–15× faster than the original KAN.

Experiment 2 (time-to-reach target MSE=0.001, up to 5000 epochs) showed RSS-KAN's higher per-iteration cost was largely offset by needing fewer iterations to converge, giving it a total convergence time competitive with the faster ReLU-KAN and SineKAN, while consistently beating the original KAN.

Ablation study (as stated in the paper)

Eight basis-function combinations were tested (Only-ReLU, Only-Sigmoid, Only-Sin, pairwise combinations, the full RSS-KAN combination, and RSS-KAN+Tanh) across four functions f_1 – f_4 . Findings reported:

- Single-basis models underperform: Only-ReLU handles piecewise/gradient-jump functions well but errs badly on strongly periodic ones; Only-Sigmoid captures smooth monotonic structure but not local oscillation; Only-Sin fits periodic functions well but converges slowly on non-periodic ones.
- The full ReLU+Sigmoid+Sin combination ("Ours") achieved the best or near-best MSE on all four test functions, e.g. 3.68×10^{-2} on f_1 versus 1.6×10^{-1} for Only-ReLU and 5.6×10^{-2} for Only-Sin.
- Adding a fourth basis function (Tanh) did not further improve results and in some cases degraded them, which the authors interpret as evidence of basis-function redundancy beyond the three chosen components.

Parameter sensitivity / dynamics analysis

Masking any one of the three combination weights (α^1 , α^2 , α^3) sharply increased error on a matched task (e.g. masking the dominant component for a given function raised MSE from the baseline $\sim 10^{-7}$ level to 10^{-2} – 10^{-1}), while masking a non-dominant component still caused measurable but smaller degradation — reported as evidence that the three basis types play complementary, not redundant, roles. A 10% perturbation to any single α produced comparatively minor error increases, indicating the learned weight allocation is robust to small perturbations while remaining sensitive to which component is entirely removed.

Real-world validation

On a multivariate meteorological time-series dataset (temperature prediction from 21 weather indicators, 10-minute sampling, 24-hour sliding window), RSS-KAN is reported to achieve lower training and validation loss than ReLU-KAN, BSRBF-KAN, Fast-KAN, Efficient-KAN, SineKAN, and a standard MLP, with predictions tracking

actual temperature trends more closely, particularly near peaks/troughs where other models showed phase lag or amplitude deviation.

Relevance to KAN-SMLM work

Not stated in the paper; report author's inference only. The matrix-only, non-recursive computational design is the most directly transferable idea — your V10/V11 pipeline uses B-spline KAN layers, and RSS-KAN's central claim is that B-splines specifically are the source of recursion-related GPU inefficiency. If localization throughput or grid-refinement speed becomes a bottleneck, a ReLU–Sigmoid–Sine hybrid basis is presented as a drop-in replacement with matrix-only computation. The high-frequency/periodic capture (Sine term) could also be relevant if PSF sidelobe structure is treated as a target for the localizer to fit, though the paper itself never discusses imaging or microscopy.

3. Wolff, Alesiani, Duhme & Jiang — Clifford Kolmogorov-Arnold Networks

arXiv:2602.05977v2 — Full text reviewed (revised June 23, 2026)

Link: <https://arxiv.org/abs/2602.05977>

What the paper proposes

CIKAN generalizes the earlier Complex-Valued KAN (CVKAN, Wolff et al., prior work) from complex numbers to arbitrary Clifford algebras $Cl(V,q)$, allowing KAN edges to learn functions $Cl \rightarrow Cl$ (e.g. quaternion-valued, or higher-grade multivector-valued) rather than just $\mathbb{R} \rightarrow \mathbb{R}$ or $\mathbb{C} \rightarrow \mathbb{C}$. Motivating examples given: electromagnetic fields (Maxwell's equations expressed compactly in $Cl_{3,0,0}$), 3D geometry, robot arm kinematics.

Core mathematical construction

For a vector space V of dimension n with quadratic form q , the Clifford algebra $Cl(V,q)$ has dimension $D = 2^n$, decomposed into grades $Cl^m(V,q)$ of dimension $C(n,m)$: grade 0 = scalars, grade 1 = vectors, grade 2 = bivectors, up to grade n = pseudoscalar. The geometric product of two vectors decomposes as $vu = v \cdot u + v \wedge u$ (scalar part + bivector part).

Two radial basis function (RBF) forms are defined for $x \in Cl$:

$$\text{naive RBF: } \phi(x) = \exp(-|x|^2) \text{ [maps } Cl \rightarrow \mathbb{R} = Cl^0(0)]$$

$$\text{Clifford RBF: } \phi_{Cl}(x) = x \cdot \tilde{x} \text{ [maps } Cl \rightarrow Cl, \text{ preserving multivector structure]}$$

where $|x|^2 = x \star \tilde{x}$ ($\tilde{x}$ = the Clifford reverse of x , $\star$ the scalar product). The paper notes the two forms are related by the gradient operator: $\phi(x)_{Cl} = -(1/2) \nabla_x \phi(x)$.

The learnable activation function on each edge is a weighted sum of these RBFs centered on a grid of points $g \in Cl$:

$$\Phi_{Cl}(x) = \sum_{g \in G} w_g \cdot \phi_{Cl}(x-g), \quad w_g \in Cl$$

A residual nonlinearity $ClSiLU(x) = w \cdot SiLU_{Cl}(x) + \beta$ (with $SiLU$ applied component-wise) is used, following the CVKAN precedent.

The curse-of-dimensionality problem and the Sobol-grid solution

A uniform ("full") grid with N_g points per dimension requires $(N_g)^D$ grid points — e.g. for quaternions ($D=4$) with $N_g=8$, this is $8^4=4096$ grid points, each with its own learnable weight $w_g \in Cl$. This exponential blow-up is the paper's stated central obstacle for higher-dimensional Clifford algebras.

The proposed fix is to replace the uniform grid with a Randomized Quasi-Monte Carlo (RQMC) scrambled Sobol sequence grid ("Sobol-CIKAN"), which the paper argues covers the hypercube $[-2,2]^D$ more evenly than either a full grid (exponentially expensive) or naive uniform-random sampling (high variance).

The paper proves (Property IV.1, Lemma IV.2, building on Owen's scrambled-net integration theory) that if grid points form a scrambled (t,m,d) -net in base b , the Sobol-grid estimator $\Phi(x) = \sum_{g \in G} w_g \cdot \phi(x-g)$ is an unbiased estimator of the integral $h(x) = \int g(y) \phi(x-y) dy$ with variance decaying as $O(n^{-1})$, where $n = |G|$ is the number of grid points — i.e. accuracy improves with more Sobol points at a standard Monte-Carlo-type rate, without requiring the grid to scale exponentially with dimension.

Batch normalization variants introduced

Three normalization schemes are proposed and compared for keeping layer outputs within grid range: dimension-wise (normalize each Clifford component across all nodes in a layer), node-wise (normalize each node's full multivector jointly across its dimensions), and component-wise (normalize each node and each dimension independently). A no-normalization baseline is also tested.

Reported experimental results

Complex-valued synthetic benchmarks (square, sin, mult, squaresquare — same tasks as the original CVKAN paper): CIKAN with a full grid matched CVKAN's performance at equal parameter count. The Sobol grid's accuracy improved with more points per dimension and matched the full grid at $N_g=8$. Table I in the source (best model per task): CIKAN achieved MSE 0.001 on square, sin, and mult, and 1.665 on squaresquare, compared to CVKAN ($lr=0.1$) at 0.001/0.001/0.005/8.150 and the improved CVKAN variant (Che et al.) at 0.009/0.005/0.029/7.355 respectively — i.e. CIKAN matched or improved on both CVKAN baselines on this benchmark set.

Holography dataset (100,000 real-world-scale samples): with the largest architecture ($3 \times 10 \times 5 \times 3 \times 1$) and node-wise batch norm, CVKAN full grid achieved test MSE 0.016 ± 0.001 ; CIKAN full grid achieved 0.025 ± 0.004 . A Sobol grid with only $N_g=7$ ($\approx 77\%$ of full-grid parameters) achieved 0.029 ± 0.004 — reported as "almost the same" as the full-grid model despite fewer parameters. Very low grid densities ($N_g=2-4$) failed to approximate the function well (MSE 0.093–1.891).

Higher-dimensional Clifford algebras (quaternion-isomorphic $Cl_{0,2,0}$; Conformal GA $Cl_{1,1,0}$; Projective GA $Cl_{1,0,1}$; Euclidean GA $Cl_{2,0,0}$), tested on square, mult, squaresquare scaled to 4D input (80,000 samples for equal sampling density): the Sobol grid matched full-grid performance at equal parameter count generally, and for $Cl_{1,0,1}$ specifically outperformed the full grid outright — e.g. on the square dataset, Sobol grid with $N_g=3$ achieved test MSE 0.004 ± 0.001 versus the full grid's 0.968 ± 0.834 (using only 1332 of 65,572 parameters, about 2%).

Batch normalization ablation: no-normalization performed far worse than all three normalization schemes on the holography dataset for both CVKAN (MSE 51.059 vs. $\sim 0.015-0.017$ for the normalized variants) and CIKAN (MSE 0.867 vs. $\sim 0.021-0.028$); among the three normalization types, no clear consistent winner emerged, and the choice "does not matter much" once some form of normalization is applied.

RBF-form ablation: the naive RBF and the Clifford RBF (CI-valued) performed similarly across most experiments — the paper reports "the way of calculating the RBFs does not seem to have a big influence on the model's performance."

Stated limitations and future work

- Training stability was reported to depend heavily on initialization for both CVKAN and CIKAN; a learning-rate increase ($0.01 \rightarrow 0.1$ relative to the earlier CVKAN paper) was needed to get competitive, more reproducible results — the paper notes instabilities can still appear across runs even with this change.
- The squaresquare task was reported as by far the hardest to fit, attributed to high output-value variability.
- The Sobol grid sometimes underperformed the full grid for specific dataset/grid-size combinations, with high standard deviation in those cases — the authors describe this as an open practical inconsistency rather than a resolved advantage.

- Explicitly flagged directions for future work: learnable RBF shape, alternative residual activations beyond SiLU, and improving training-stability robustness to initialization.

Relevance to KAN-SMLM work

Not discussed in the paper; report author's inference only. The most plausible connection is structural rather than direct: your V10_fixed LayerNorm fix for mode collapse is conceptually adjacent to this paper's node-wise vs. dimension-wise vs. component-wise batch-norm comparison — the paper's finding that normalization type matters much less than the presence of normalization itself may be a useful data point if further normalization tuning is considered. The Clifford-algebra generalization itself (mapping learnable KAN edges to multivector-valued outputs) is not obviously applicable to 2D/3D coordinate regression in SMLM unless DH-PSF orientation/phase information were explicitly encoded as a Clifford-valued quantity, which the paper does not suggest and which would be a substantial reinterpretation, not something drawn from the text.

Summary Table

Paper	Venue / Date	Core contribution	Confidence
Faroughi et al. — KAN review & synthesis	Neural Networks, Mar 2026	Reviews data-driven/PIKAN/DeepOKAN literature; new NTK & accuracy benchmarks vs. MLP/PINN/DeepONet	
Zhang, Fan & Lu — RSS-KAN	Neurocomputing, Jan 2026	ReLU+Sigmoid+Sine hybrid basis; matrix-only computation; 2–3 order-of-magnitude accuracy gain, 5–15x speedup vs. original KAN	
Wolff et al. — Clifford KAN	arXiv, rev. Jun 2026	Extends KAN to Clifford-algebra-valued (multivector) edges; Sobol-sequence grid to avoid exponential parameter growth	

All three entries above are now full-text reviewed (upgraded from the prior report's abstract-only status).

Notes on This Report

- All figures, equations, and quoted numerical results above are transcribed directly from the uploaded PDFs' text and tables — nothing is estimated or extrapolated beyond what the source states, except where explicitly marked as "report author's inference."
- Equations are rendered as italicized plain text (no native LaTeX renderer in this document pipeline); the original PDFs remain the authoritative source if exact notation is needed for citation.
- Ready to be appended to the original "KAN Architecture Research Report 2026" document.